Intermediate band mobility requirements in intermediate band solar cells

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***Abstract*—The term "intermediate band solar cells" invokes the idea of band transport in the intermediate band (IB), but since current is not extracted directly from the IB, it is not clear whether transport within the IB is required for high efficiency. We propose a simple rule to understand the requirements for IB transport in efficient devices. If, at each point in the IB, the generation rate of electrons optically excited from the valence band to the IB equals the generation rate of electrons moved to the conduction band from the IB, we call the device locally current matched (LCM). In the LCM case, IB transport is not required for high efficiency. When a device does not have local current matching, it must have IB transport or else recombination will decrease efficiency. Using the coupled Poisson/drift-diffusion model Simudo, we quantify the requirements for IB mobility in a range of cases. We verify that for LCM devices, IB mobility is not important. For mismatched devices, modest values of IB mobility near 1-30 cm^2/Vs are sufficient to attain high efficiency even in the absence of local current matching. We also show that device design can ameliorate the efficiency decrease from using locally mismatched materials with low mobility; choosing an appropriate polarity of the p-IB-n junctions can significantly restore efficiency even with low IB mobility. These results provide valuable guidance to developers of IB materials and devices, showing that IB mobilities are important when the subgap transition strengths differ significantly.**

***Index Terms*—photovoltaic cells, mathematical model, intermediate band materials.**

I. Introduction

Intermediate band solar cells (IBSCs) hold the promise of providing the high efficiency of multijunction solar cells with simpler device structures and growth requirements [1], [2]. Intermediate band (IB) materials are like semiconductors except with an extra band of allowed electronic energy levels deep inside the gap, allowing subgap light absorption from valence band (VB) to IB and IB to conduction band (CB). While the theoretical potential of IBSCs has been well described by detailed balance studies [1], [3], the material requirements for making effective IBSCs remain less well understood. Detailed balance studies consider idealized materials with infinite carrier mobilities and only radiative recombination [4]. Real intermediate band (IB) devices generally have short nonradiative lifetimes [5]–[8], so clear guidelines are essential to determine whether material properties are sufficient to enable high-efficiency devices. We have previously described a figure of merit ν, which captures the tradeoff between subgap light absorption and carrier collection in IBSCs, focusing on the CB and VB carrier transport and lifetime [6], [9], [10].

In this work, we show the circumstances under which an IBSC requires carrier mobility in the IB itself for efficient operation. Devices that have local current matching of the subgap generation currents can be highly efficient even without IB mobility μ_I. If one of the subgap optical transitions is much stronger than the other, as commonly occurs, absorption is not locally matched throughout the device, and μ_I must be large enough for the electrons to move between the locations where the subgap absorptions occur. Quantitative requirements for IB mobility μ_I in IBSCs have rarely been discussed, as the tools to study IB transport have largely not been available. In Ref. [10], we found that when the CB and VB properties are equal and therefore there is approximate local current matching, μ_I has no effect on efficiency. Reference [11] showed that μ_I can affect the short circuit current when the subgap absorptivities have different strengths. Here, we study the importance of μ_I for efficient devices. We use Simudo, a coupled Poisson/drift-diffusion (PDD) model that can include intermediate bands, including the self-consistent treatment of subgap absorptions, which change due to photofilling of the IB [11]. Simudo includes transport in all three bands of the material, allowing determination of the effects of mobility in the IB. In contrast, detailed balance models and those built from detailed balance [12] consider infinite mobility and cannot treat such effects. Other PDD models are capable of performing this study but have not done so [13], [14].

II. The importance of IB mobility

In steady-state operation, at each point in the device the number of electrons entering the IB must equal the number exiting, including all optical generation, recombination, and transport processes. It is helpful to begin by analyzing the optical generation processes only. The IB material is characterized by its band-edge energies E_k, where k can be C, V, or I, for the CB, VB, and IB, respectively. Consider a material with non-overlapping absorptions, where for each photon frequency ω, photons can be absorbed only by the highest-energy transition consistent with energy conservation [1]. We consider $E_{IV} < E_g/2$, where $E_{XY} \equiv E_X - E_Y$ and $E_g = E_{CV}$ is the bandgap. We define $\omega_{XY} \equiv E_{XY}/\hbar$, where $\hbar$ is Planck's constant. Given incident spectral photon flux $\Phi_0(\omega)$, the absorbable photon flux for transitions CI and IV are then

$$\dot{N}_{IV} = \int_{\omega_{IV}}^{\omega_{CI}} \Phi_0(\omega)\mathrm{d}\omega \tag{1}$$

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$$\dot{N}_{CI} = \int_{\omega_{CI}}^{\omega_g} \Phi_0(\omega)\mathrm{d}\omega.$$

While simulations in this manuscript use the non-overlapping absorption assumption, we can define $\dot{N}_{XY}$ for a general material with subgap absorption coefficients $\alpha_{XY}(\omega)$, where XY is either CI or IV. In this case, we can consider an infinitely thick device, in which photons of frequency ω are divided between the two transitions proportionally to $\alpha_{XY}(\omega)$. Then

$$\dot{N}_{IV} = \int_0^\infty \frac{\alpha_{IV}(\omega)}{\alpha_{CI}(\omega) + \alpha_{IV}(\omega)} \Phi_0(\omega)\mathrm{d}\omega \tag{2}$$

$$\dot{N}_{CI} = \int_0^\infty \frac{\alpha_{CI}(\omega)}{\alpha_{CI}(\omega) + \alpha_{IV}(\omega)} \Phi_0(\omega)\mathrm{d}\omega,$$

where we take $\alpha_{XY}(\omega)/\left(\alpha_{CI}(\omega) + \alpha_{IV}(\omega)\right)$ to be zero at frequencies where both α_{XY} are zero. These expressions require that $\alpha_{XY}(\omega)$ be considered spatially constant in the IB region, so could be evaluated at equilibrium. They reduce to Eq. 1 in the case of nonoverlapping absorptions.

The optimal detailed-balance efficiency occurs when the subgap transitions have equal photon fluxes, that is, when $\dot{N}_{CI} = \dot{N}_{IV}$. We define the global current mismatch m_G,

$$m_G = \frac{\left|\dot{N}_{CI} - \dot{N}_{IV}\right|}{\dot{N}_{CI} + \dot{N}_{IV}}. \tag{3}$$

With non-overlapping absorptions, m_G is a property of the spectrum and energy levels of the IB material alone and does not depend on the magnitude of $\alpha_{XY}(\omega)$ in the frequency domain where it is nonzero.

We also consider the local subgap absorptions. At each point x, the rates of electrons entering and exiting the IB due to the IV and CI absorption processes (per unit volume) are

limits need to be swapped for this case, with E_IV<Eg/2

$$g_{IV}(x) = \int_{\omega_{IV}}^{\omega_{CV}} \Phi(\omega, x)\alpha_{IV}(\omega, x)\mathrm{d}\omega \tag{4}$$

$$g_{CI}(x) = \int_{\omega_{CI}}^{\omega_{IV}} \Phi(\omega, x)\alpha_{CI}(\omega, x)\mathrm{d}\omega, \tag{5}$$

where $\Phi(\omega, x)$ is the spectral photon flux at position x and α_{XY} is the absorption coefficient between bands X, Y. We can characterize the scale of the local mismatch through the device by defining the integrated local mismatch factor m_L,

$$m_L \equiv \frac{\int \left(g_{IV}(x) - g_{CI}(x)\right)^2 \mathrm{d}x}{\int \left(g_{IV}(x) + g_{CI}(x)\right)^2 \mathrm{d}x}. \tag{6}$$

This local mismatch can be large even when $m_G = 0$, if the subgap absorption strengths differ. Note that m_L can also be zero even when m_G is finite, if the device does not absorb all incident photons.

The subgap absorption coefficients α_{XY} depend on the IB filling fraction f_I. When the IB is full, the IV transition cannot occur, and when it is empty the CI transition cannot occur. We parametrize α_{XY} using a notation where the IB is formed from defects with concentration N_I. We then write

$$\alpha_{IV}(\omega) = N_I \sigma_{IV}^{\mathrm{opt}}(\omega)(1 - f_I) \tag{7}$$

$$\alpha_{CI}(\omega) = N_I \sigma_{CI}^{\mathrm{opt}}(\omega) f_I, \tag{8}$$

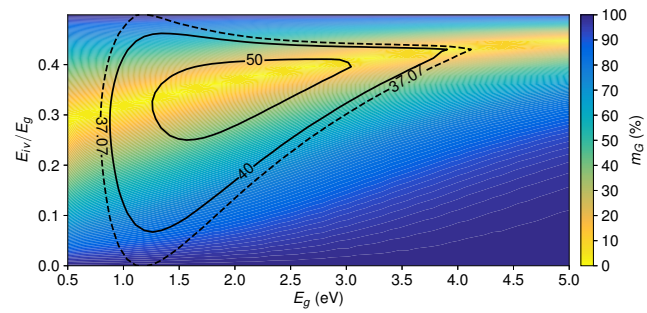

Fig. 1. Global current mismatch m_G for intermediate band absorber with non-overlapping absorptions with a 6000 K black body spectrum. Contours show detailed balance efficiencies [1] with concentration factor $X = 1000$. The dashed contour shows the single-junction detailed balance limit [4] at $X = 1000$ of 37.07%. Achieving the highest efficiencies generally requires $m_G < 50\%$.

where $\sigma_{XY}^{\mathrm{opt}}$ is an optical capture cross section. The dependence of α_{XY} on f_I applies to all IB systems, not just those formed from defects [15]. The local mismatch m_L changes with f_I, and therefore depends on both incident radiation and applied voltage.

When $m_L \neq 0$, keeping the IB in steady state requires that there either be transport in the IB or else recombination processes to balance the (locally) mismatched currents. Those recombination processes can represent loss for the device. If the system has $m_L = 0$, then there is no need for IB transport to attain high efficiency. Conversely, a system with sufficiently high mobility can be highly efficient with large m_L.

Considering a family of IB materials with fixed E_g, m_G changes as E_{IV} changes. As m_G increases from zero with fixed E_g, the detailed balance efficiency limit for the IBSC, which gives the maximum attainable efficiency for a purely radiative device, decreases. For $m_G \gtrsim 50\%$, the IBSC is no longer able to achieve efficiencies significantly higher than single-junction devices, as shown in Fig. 1 at $X = 1000$ suns concentration. We therefore limit our study to $m_G \leq 50\%$. Realistic values of m_L for IBSC devices, however, range nearly up to 100%.

Some deviation from global current matching is clearly acceptable without harm to device efficiency; in Ref. [10], we found no effects of μ_I on device efficiency for a system with $\sigma_{CI}^{\mathrm{opt}} = \sigma_{IV}^{\mathrm{opt}}$, even though $m_G = 18.3\%$. In that study, electron and hole properties were chosen to be equal, so the only asymmetry arose from m_G. As we show in more detail below, local subgap absorption strengths being matched is more important than the global current match in determining whether IB mobility is required for device efficiency.

III. Simulation Method

The full capabilities of Simudo are detailed in Ref. [11], and we give a brief overview of the points relevant to this study. We consider three bands k, where k can be C, V or I, for the CB, VB, and IB, respectively, each of which is characterized by a band-edge energy E_k and an effective density of states N_k. For

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convenience, we treat the IB as having a well-defined energy E_I, though actual IBs have a nonzero density of states over a range of energies [11]. The CB and VB charge densities, n and p, respectively, are treated in the Boltzmann approximation while degenerate statistics are used for the IB filling fraction f_I. The electrostatic potential ϕ is connected to the charge density ρ through Poisson's equation,

$$\rho = q[-n + p - N_I(f_I - f_{I,0}) + N_D - N_A], \tag{9}$$

where q is the fundamental charge, and we include fixed background charge densities N_D for donors and N_A for acceptors, and $f_{I,0}$ is the equilibrium filling of the IB at $T = 0$. The current density in each band k is proportional to its mobility μ_k and includes both drift and diffusion correctly for both non-degenerate and degenerate bands [14].

We treat $\Phi_0(\omega)$ from a black body spectrum at 6000 K normally incident on the device. The optical absorption at position x obeys

$$\frac{\mathrm{d}\Phi(\omega, x)}{\mathrm{d}x} = -\alpha(\omega, x)\Phi(\omega, x) \tag{10}$$

where the absorption coefficient $\alpha = \alpha_{CV} + \alpha_{CI} + \alpha_{IV}$. While α_{CV} is a material parameter, the subgap transitions depend on f_I as shown in Eq. 8. In this work, we consider non-overlapping absorptions and set σ^{opt}_{XY} and α_{CV} to be constant over their allowed range of ω.

TABLE I
PARAMETERS FOR DEVICE SIMULATIONS, ADAPTED FROM [16]

Value	Definition
$E_g = 1.67$ eV	Conduction band edge energy
$N_C = N_V = 5 \times 10^{18}$ cm^{-3}	CB and VB effective density of states
$N_I = 10^{17}$ cm^{-3}	IB density of states
$\mu_C = \mu_V = 2000$ cm^2/Vs	CB and VB mobility
$\mu_I = 0.001 - 100$ cm^2/Vs	IB mobility
$v^{\text{th}} = 2 \times 10^7$ cm/s	Thermal velocity in CB and VB
$\sigma^{\text{th}} = 1.2 \times 10^{-15}$ cm^2	Thermal cross section for CI and IV
$\alpha_{CV} = 10^4$ cm^{-1}	Absorption coefficient for CV process
$\sigma^{\text{opt}}_{CI} = (1-5) \times 10^{-12}$ cm^2	Optical cross section for CI process
$\sigma^{\text{opt}}_{IV} = (1-5) \times 10^{-12}$ cm^2	Optical cross section for IV process
$f_{I,0} = 1/2$	Charge-neutral IB filling fraction
$T_c = 300$ K	Cell temperature
$T_s = 6000$ K	Sun temperature
$X = 1000$	Solar concentration
$w_{\text{p}} = 1$ μm	p region length
$w_{\text{n}} = 1$ μm	n region length
$N_A = 10^{17}$ cm^{-3}	p-type doping
$N_D = 10^{17}$ cm^{-3}	n-type doping
$\varepsilon_r = 13$	Relative permittivity

TABLE II
E_{IV} AND ASSOCIATED m_G UNDER 6000 K BLACK BODY SPECTRUM WITH $E_g = 1.67$ eV

E_{IV}	m_G
0.595 eV	1.7%
0.561 eV	8.3%
0.503 eV	23.3%
0.384 eV	48.4%

We study a p-IB-n model system, similar to that studied in Refs. [11], [16], with parameters shown in Table I. Figure 2 shows a band diagram for an IBSC device with low μ_I. We consider a sufficiently high conductivity IB region that there are two depletion regions rather than a p-i-n structure, which can be seen in the approximately flat-band condition in the middle of the IB region in Fig. 2. We use a 6000 K black body spectrum under $X = 1000$ concentration and adjust $E_{IV} < E_g/2$ to achieve four values of m_G, all with an excess of photons for the IV process, shown in Table II. Our choice of $E_g = 1.67$ eV is purposely not optimal for detailed balance efficiency for IBSCs; we instead seek generic behaviors for a representative but not perfect IB material. There is a 50 nm front surface field region with doping $N_A = 10^{19}$ cm^{-3} at the front, and we set the surface recombination velocity at the contacts for minority carriers to zero.

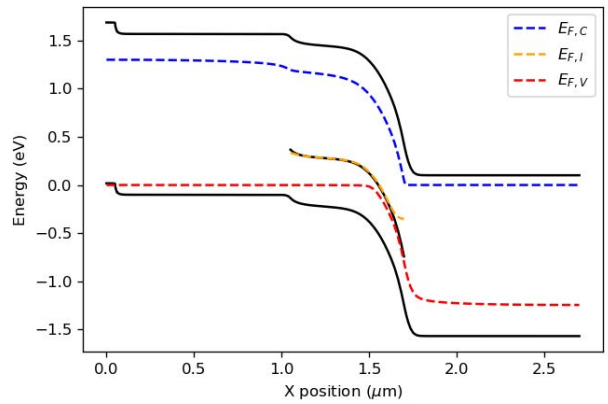

Fig. 2. Band energy diagram for a p-IB-n IBSC at short circuit under 1000 suns black body illumination with parameters as in Table I, $\mu_I = 0.001$ cm^2/Vs, $m_G = 0.23$, and $\sigma^{\text{opt}}_{CI} = \sigma^{\text{opt}}_{IV} = 10^{-12}$ cm^{-2}. Black lines show, from top to bottom, the conduction, intermediate, and valence band energies, respectively. Dashed lines show the quasi-Fermi levels for each band.

We consider IB materials of high quality. We have shown previously that exceeding the Shockley-Queisser limit with $E_g = 1.67$ eV at $X = 1$ requires a figure of merit

$$\nu = \frac{kT_c}{q} (\mu\tau)_{\text{min}} \alpha^2_{\text{min}}$$

greater than 50, where μ is the mobility of CB and VB carriers, τ is the lifetime of those carriers, α is the subgap absorption coefficient, q is the fundamental charge, and we take the smallest value for $\mu\tau$ and α for the two bands [10]. We introduce Shockley-Read trapping from CB to IB and from IB to VB [11], [17] with identical rates for both processes. The thermal velocity v^{th} and capture cross section σ^{th} in Table I give $\nu = 104$, which is a high quality device that is still far from the radiative limit. These values give a diffusion length of electrons and holes in the IB region of approximately 2 μm. Note that ν does not depend on the IB mobility.

IV. RESULTS

We simulate these devices using Simudo, optimizing the width of the IB region w_I for each set of material parameters

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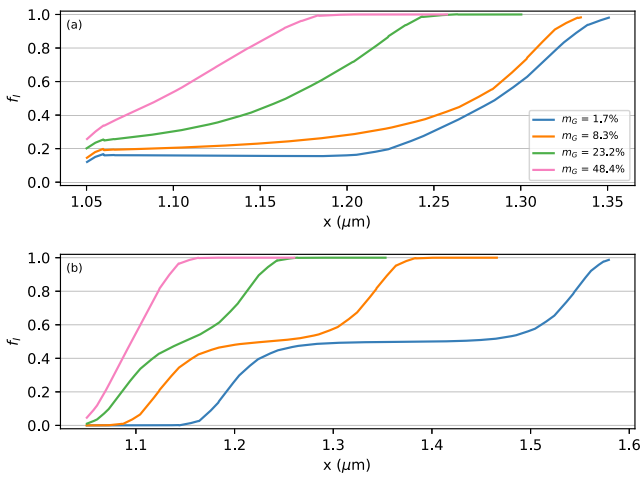

Fig. 3. Intermediate band filling fraction at short circuit under illumination for IBSC's with $r^{\text{opt}} = 1/5$. Panel (a) has $\mu_I = 0.001$ cm^2/Vs and (b) has $\mu = 100$ cm^2/Vs. Four values of m_G are shown. Each of these eight devices has w_I chosen to optimize efficiency.

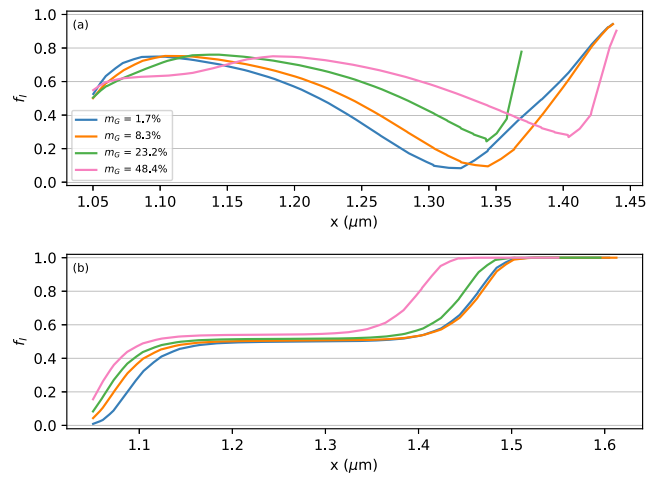

Fig. 4. Intermediate band filling fraction at short circuit as in Fig. 3 with $r^{\text{opt}} = 5$.

to obtain the highest efficiency. We vary m_G by changing E_{IV} and vary m_L by changing the optical cross section ratio

$$r^{\text{opt}} \equiv \frac{\sigma_{IV}^{\text{opt}}}{\sigma_{CI}^{\text{opt}}}. \tag{11}$$

Figure 2 was obtained after optimizing for w_I, which is 0.65 μm in that case. Table III gives the resulting optimized efficiencies and Table IV gives the optimized w_I. We observe further decrease of efficiency of less than 0.1% (absolute) when μ_I is made smaller than 0.001 cm^2/Vs. Table III shows that μ_I of 1 cm^2/Vs is sufficient to achieve high efficiency in all studied cases. Reducing μ_I to 0.001 cm^2/Vs, however, significantly decreases efficiency when $r^{\text{opt}} = 5$, with a smaller decrease when $r^{\text{opt}} = 1/5$.

A material with high μ_I can redistribute IB electrons regardless of the spatial location of each subgap absorption, so high efficiency can be achieved for essentially any value of r^{opt}. Even when r^{opt} is not 1, photofilling effects tend to reduce m_L in materials with low μ_I by changing f_I. For example, if $r^{\text{opt}} = 1/5$ and $m_G = 0$, the initially strong CI absorption depletes the IB, which brings the subgap generations closer to alignment. They can be made equal if $f_I = 1/6$, as can be seen from Eq. 8. Figure 3 shows f_I through the IB region for devices with low mobility (a) and high mobility (b), with a range of m_G values. The blue line in panel (a), with m_G close to 0, indeed begins with f_I close to 1/6. As m_G increases and there are excess photons for the IV transition, local subgap current matching is achieved at higher values of f_I, as shown in the other curves. In contrast, in the high-mobility case, f_I is close to zero at the front of the IB region, producing an excess of IV transitions. The resulting electrons move through the IB to the back of the IB region, where the CI transition occurs.

When $r^{\text{opt}} = 5$, establishing local current matching requires $f_I = 5/6$. At the front of the IB region, where the light intensity is strongest, however, there is a p-IB junction. Near that junction, f_I is forced to be small, due to the electric field pushing electrons out of the depletion region. Figure 4 shows f_I for devices with $r^{\text{opt}} = 5$, with panel (a) again showing the low mobility case. Note that f_I is over 0.5 at the front of those devices, but they do not reach the 5/6 level required for local current matching. As a result, there is extra recombination, and these devices have lower efficiencies than those in Fig. 3(a), as shown in Table III. Deeper in the IB region, f_I decreases as the photons available for the IV transition are depleted, and the IB is photodepleted by the CI transition. The reduced efficiency in this configuration suggests a simple method to increase IB device efficiency in the case that one has an IB with r^{opt} significantly different from 1 and small μ_I. If $r^{\text{opt}} > 1$, use an n-IB-p structure in place of the p-IB-n, so the depletion region f_I profile matches better the requirements for equal local subgap generations.

In contrast, Figure 4(b) shows that at high μ_I, the f_I curves are all similar in character to the $m_G = 1.7\%$ case in Fig. 3. At the front of the IB region, f_I is small, giving an overwhelming excess of IV transitions over CI transitions in all cases. The electrons transport through the IB region to the middle and rear of the device, where the CI absorption process occurs.

In both Figures 3 and 4 note that when μ_I is small, changing m_G has a smaller effect on the qualitative shape of $f_I(x)$ than changing r^{opt} has. In contrast, when μ_I is higher, f_I has a similar shape in all devices, showing the formation of depletion regions at the junctions with the p- and n-type regions. The photofilling effects, raising f_I in the interior of the device above 0.5, are due to m_G and are stronger in Fig. 3(b) than in Fig. 4(b). This observation is consistent with r^{opt} having a larger impact than m_G on device efficiency at low μ_I.

Figure 5 shows that at low mobility, efficiency is negatively

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TABLE III

OPTIMIZED EFFICIENCIES FOR DEVICES WITH VARYING m_G AND μ_I. WHEN $\frac{\sigma_{IV}^{\text{opt}}}{\sigma_{CI}^{\text{opt}}} = 1$, BOTH CROSS SECTIONS ARE 10^{-12} CM^2. FOR THE OTHER RATIOS, THE SMALLER CROSS SECTION REMAINS AT 10^{-12} CM^2.

		μ_I (cm^2/Vs)			
m_G	$\frac{\sigma_{IV}^{\text{opt}}}{\sigma_{CI}^{\text{opt}}}$	0.001	1	30	100
1.7%	5	33.97	41.00	41.39	41.39
	1	39.70	41.00	41.00	41.00
	1/5	38.40	40.84	40.98	40.98
8.3%	5	34.14	40.73	41.02	41.02
	1	40.18	40.89	40.86	40.86
	1/5	39.63	41.37	41.40	41.40
23.3%	5	33.97	39.30	39.55	39.54
	1	39.14	39.49	39.45	39.45
	1/5	40.10	40.45	40.44	40.44
48.4%	5	32.50	35.63	35.88	35.87
	1	35.59	35.81	35.78	35.78
	1/5	36.81	36.77	36.76	36.76

TABLE IV

OPTIMIZED IB REGION WIDTH w_I WITH THE SAME PARAMETERS AS IN TABLE III.

		μ_I (cm^2/Vs)			
m_G	$\frac{\sigma_{IV}^{\text{opt}}}{\sigma_{CI}^{\text{opt}}}$	0.001	1	30	100
1.7%	5	0.39	0.55	0.56	0.56
	1	0.69	0.67	0.64	0.64
	1/5	0.30	0.50	0.53	0.53
8.3%	5	0.39	0.57	0.56	0.56
	1	0.68	0.62	0.60	0.60
	1/5	0.28	0.40	0.42	0.42
23.3%	5	0.32	0.56	0.55	0.55
	1	0.65	0.60	0.57	0.57
	1/5	0.25	0.30	0.30	0.30
48.4%	5	0.23	0.53	0.50	0.50
	1	0.59	0.55	0.53	0.53
	1/5	0.21	0.21	0.21	0.21

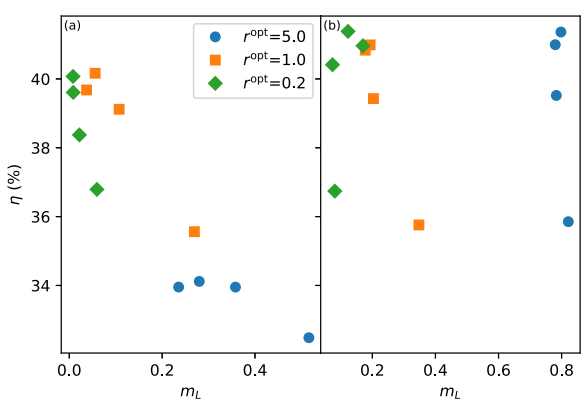

Fig. 5. Optimized device efficiency η and local mismatch m_L at the maximum power point voltage with (a) $\mu_I = 0.001$ cm^2/Vs and (b) $\mu_I = 100$ cm^2/Vs. The colours indicate $r^{\text{opt}} = \sigma_{IV}^{\text{opt}}/\sigma_{CI}^{\text{opt}}$. At low mobility, m_L is negatively correlated with efficiency, while there is no relationship at higher mobility. The multiple points of each color are the four values of m_G shown in Table III.

correlated with m_L at small μ_I, evaluated at the maximum power point voltage. At higher mobility, there is no strong relationship between m_L and efficiency. Evaluating m_L at the operating voltage displays the importance of local current matching for device efficiency in devices with low μ_I. The same relationship between m_L and η holds at short circuit or at equilibrium when $r^{\text{opt}} = 1$, but at the other studied values of r^{opt}, m_L at short circuit does not have a clear relationship to efficiency (not shown). The IB filling fraction changes sufficiently with voltage and illumination that m_L in these other conditions is not indicative of device efficiency.

In summary, if an IB material has sufficiently high IB mobility, it can achieve high efficiency regardless of whether its subgap absorptions are locally matched. Materials with equal-strength subgap absorptions maintain local subgap current matching and can achieve high efficiencies even without significant conduction in the IB. As materials deviate from local current matching, however, IB mobility becomes more important for achieving the high efficiency potential of IBSCs. Though balance between the subgap photon fluxes is important for detailed balance efficiencies, in the cases studied here the subgap absorption strengths are more important for understanding whether IB mobility is important in a candidate device. IB materials with larger IV than CI absorption coefficients can reduce the need for IB mobility by being in an n-IB-p structure, while a p-IB-n structure is best for the case of larger CI than IV absorption coefficients. These results provide valuable experimental guidance for members of the IBSC community, who can readily determine their IB energy levels and subgap absorption coefficients. With that information, they can determine whether it is important to engineer IB mobility (e.g., through miniband formation, in the case of quantum dot IBSCs). In the cases studied here, quite modest values of $\mu_I \approx 1$ cm^2/Vs are sufficient to allow high efficiency. IB mobilities are rarely measured [18], but they are important quantities for systems with significantly mismatched absorptions.

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